

Trilaterales Transferprojekt DFG - Fraunhofer

December 6, 2023



Agenda

1. Introduction round
2. Organization
3. Deliverables & work packages

Introduction round

Organization

Deliverables

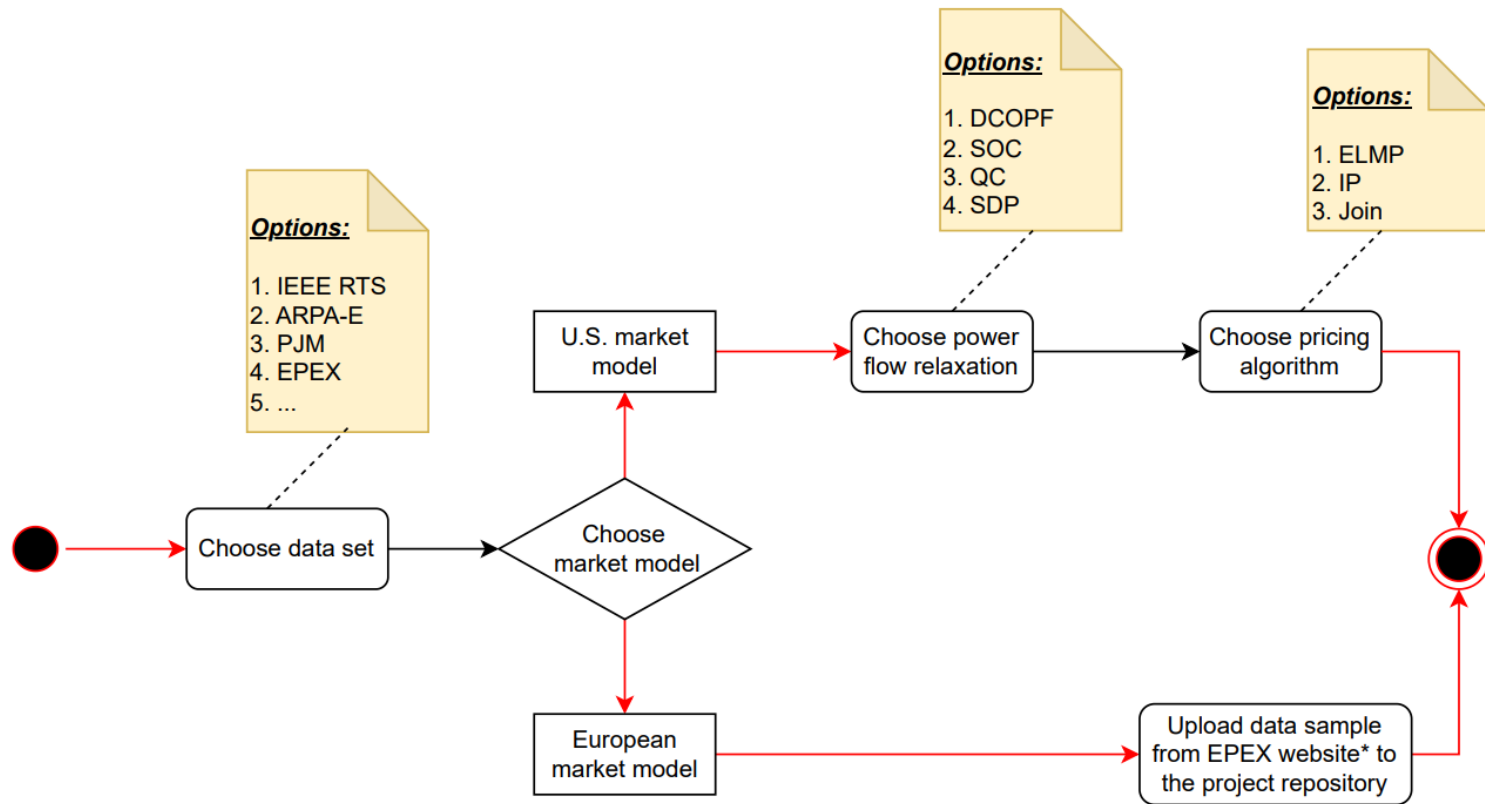
- **Simulator** that allows running simulations with different **pricing rules**, different **relaxations of ACOPF**, and different **data sets** of realistic problem size (backend implementation in Python).
- Detailed documentation of the simulator's functionalities.
- Descriptions of the implemented mathematical models.

Next: user flow

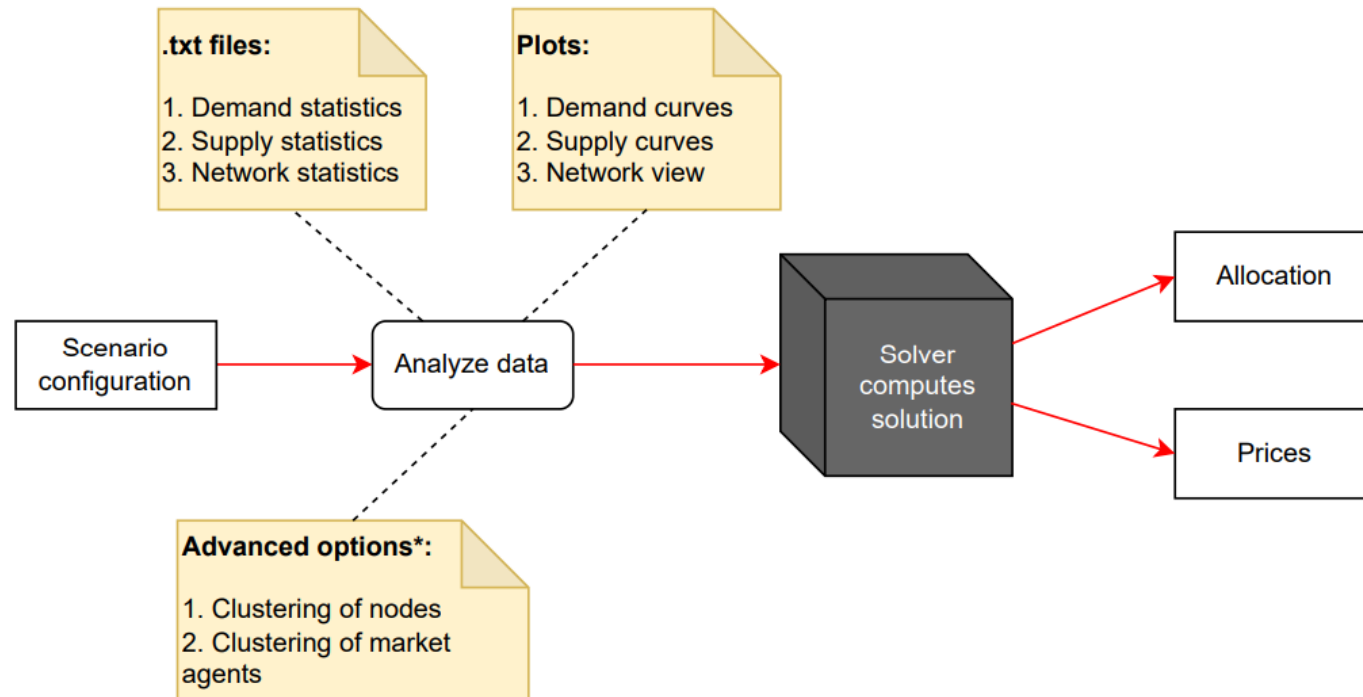
Assumptions

1. User has cloned the project repository on their local machine and followed the installation instructions.
2. User has access to an IDE (VSCode, PyCharm, etc.).
3. User has basic Python knowledge.

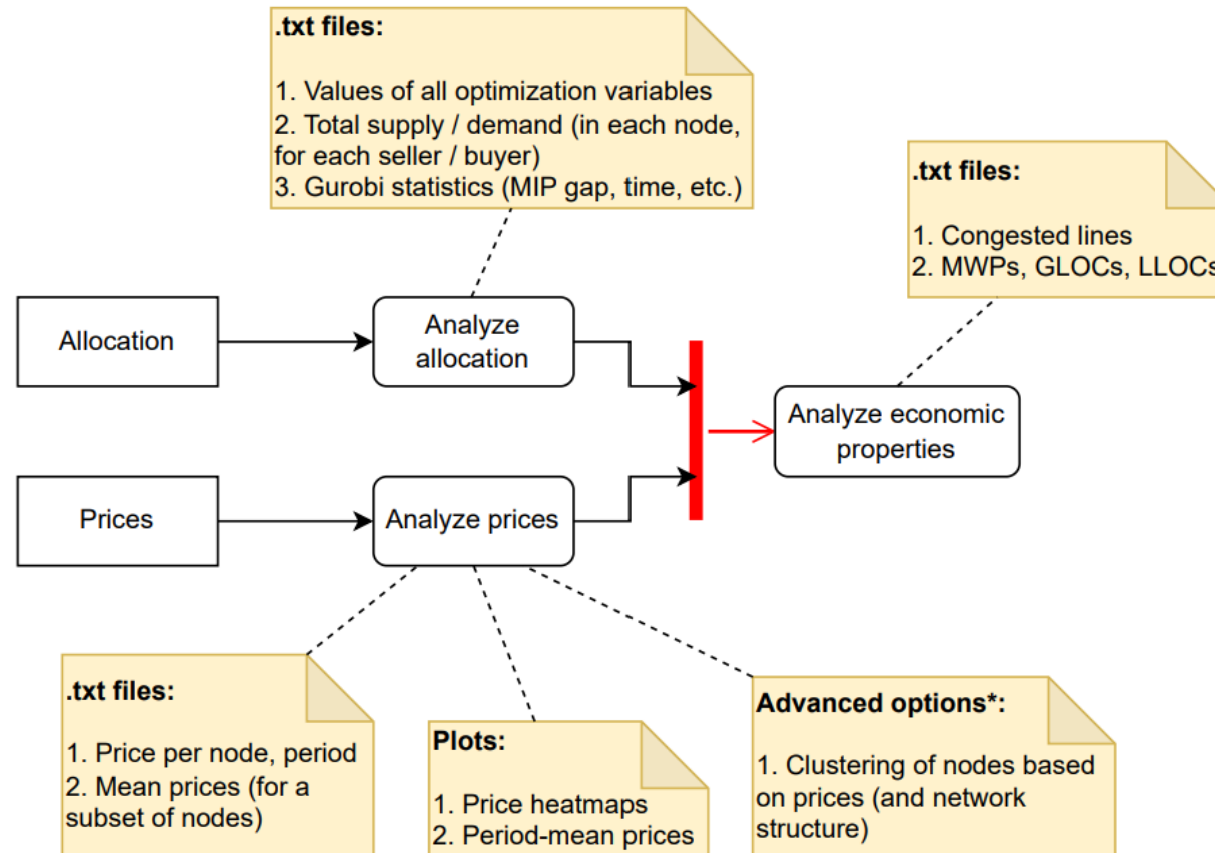
Step 1: Configure a scenario in the Python code



Step 2: Let the solver compute an allocation and prices



Step 3: Analyze allocation and prices



No *perfect* dataset

- There is no available dataset that allows analyzing both the U.S. and the European market models.
- EPEX dataset: block and hourly bids, but **no transmission network information** (e.g., cross-border capacities)
- BZR: expand with **demand-side bids (hourly + block)**

Work packages

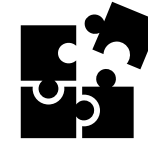
Work package (WP)	Year 1	Year 2	Year 3	TUM	FIT	TenneT
WP 1: Framework design	**			2 PM	2 PM	2 PM
WP 2: Preparation of realistic problem instances	****			3 PM	5 PM	1 PM
WP 3: Development of linear approximations (DCOPF, zonal)	****	**		4 PM	2 PM	0 PM
WP 4: Development of convex relaxations (SOC, QC, SDP)		****		4 PM	3 PM	0 PM
WP 5: Development of IP pricing and ELMP pricing rules		****		7 PM	3 PM	1 PM
WP 6: Development of the Join price rule		**	**	6 PM	4 PM	1 PM
WP 7: Development of a framework for regression testing and numerical experiments	**	****		2 PM	5 PM	1 PM
WP 8: Experimental evaluation of different pricing rules			**	2 PM	6 PM	3 PM

Stars indicate the quarters of a year

WP 1: Framework design

Goal: define framework functionalities & architecture

Deadline: May 2024



Tasks:

1. Decide on the modeling scope of market participants and transmission network
2. Define functional requirements, components, etc.

Work package (WP)	Year 1	Year 2	Year 3	TUM	FIT	TenneT
WP 1	**			<i>2 PM</i>	<i>2 PM</i>	<i>2 PM</i>

WP 2: Preparation of realistic problem instances

Goal: mimic real data used in European electricity markets

Deadline: December 2024



Tasks:

1. Analyze IEEE RTS, ARPA-E, PJM, EPEX datasets
2. Based on the insights obtained from the previous task, construct instances that resemble European electricity markets
3. Implement procedures for data parsing, cleaning and validation

Work package (WP)	Year 1	Year 2	Year 3	TUM	FIT	TenneT
WP 2	****			3 PM	5 PM	1 PM

WP 3: Development of linear approximations (DCOPF, zonal)

Goal: implement nodal and zonal market clearing mechanism

Deadline: May 2025



Tasks:

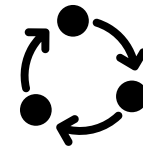
1. Implement the DCOPF model for the nodal market design
2. Replicate the EUPHEMIA algorithm for the zonal market design
3. Conduct initial experiments based on the data from WP 2

Work package (WP)	Year 1	Year 2	Year 3	TUM	FIT	TenneT
WP 3	****	**		<i>4 PM</i>	<i>2 PM</i>	<i>0 PM</i>

WP 4: Development of convex relaxations (SOC, QC, SDP)

Goal: benchmark different power flow models w.r.t. their underlying market clearing

Deadline: December 2025



Tasks:

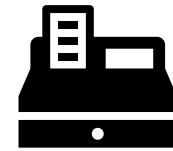
1. Implement power flow relaxations
2. Integrate functionalities from the PowerModels library

Work package (WP)	Year 1	Year 2	Year 3	TUM	FIT	TenneT
WP 4		****		<i>4 PM</i>	<i>3 PM</i>	<i>0 PM</i>

WP 5: Development of IP pricing and ELMP pricing rules

Goal: analyze existing pricing rules

Deadline: December 2025



Tasks:

1. Implement IP and ELMP pricing rules
2. Conduct experiments and analyze the behavior of the pricing rules under the power flow models developed in WP 4

Work package (WP)	Year 1	Year 2	Year 3	TUM	FIT	TenneT
WP 5		****		<i>7 PM</i>	<i>3 PM</i>	<i>1 PM</i>

WP 6: Development of the Join pricing rule

Goal: analyze pricing rules that address the downsides of traditional pricing approaches

Deadline: May 2026



Tasks:

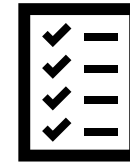
1. Extend the existing formulation of the Join pricing rule for the power flow models developed in WP 4
2. Implement other recently developed pricing rules

Work package (WP)	Year 1	Year 2	Year 3	TUM	FIT	TenneT
WP 6		**	**	<i>6 PM</i>	<i>4 PM</i>	<i>1 PM</i>

WP 7: Development of a framework for regression testing and numerical experiments

Goal: implement a software quality assurance plan

Deadline: December 2026



Tasks:

1. Develop tests to ensure the quality of the framework
2. Create test cases that have external validity
3. Draft an experimental design that will be extended in WP 8

Work package (WP)	Year 1	Year 2	Year 3	TUM	FIT	TenneT
WP 7	**	****		2 PM	5 PM	1 PM

WP 8: Experimental evaluation of different pricing rules

Goal: analyze allocation and pricing results, test the capabilities of the framework

Deadline: May 2026



Tasks:

1. Conduct extensive numerical experiments
2. Analyze the scalability of the implemented algorithms and the properties of the resulting allocation and prices for real-world power networks

Work package (WP)	Year 1	Year 2	Year 3	TUM	FIT	TenneT
WP 8			**	2 PM	6 PM	3 PM

Repository & meetings

- All files will be stored in a Github repository.
- Biweekly/monthly meetings with the operational team.
- Bimonthly meetings with the whole team.